

HART ISLAND, NS, Canada

WMO: 714190

Lat: 45.344N

Lon: 60.990W

Elev: 8

StdP: 101.23

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.0	-11.8	-19.5	0.7	-12.5	-17.2	0.8	-10.8	19.6	-4.3	18.4	-3.6	8.6	290	0.811

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	23.5	19.0	22.2	18.5	21.1	18.1	20.6	22.0	19.8	21.2	19.1	20.4	5.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	14.8	21.2	19.3	14.1	20.6	18.5	13.4	19.8	59.4	22.0	56.9	21.1	54.3	20.3	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	15.0	13.3	DB	-17.5	27.0	2.6	1.9	-19.4	28.3	-20.9	29.4	-22.3	30.5	-24.2	31.9	(4)
(5)			WB	-18.1	21.9	2.6	1.3	-19.9	22.9	-21.4	23.7	-22.9	24.4	-24.7	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-2.8	-3.3	-1.0	2.9	7.4	12.1	16.9	18.6	15.9	10.6	5.6	0.8
	DBStd	8.15	4.55	4.25	3.41	2.52	2.66	2.80	2.19	1.71	2.60	3.04	3.63	4.04
	HDD10.0	1874	396	372	340	213	91	11	0	0	0	29	137	286
	HDD18.3	4162	654	605	598	463	341	187	54	17	80	239	381	544
	CDD10.0	799	0	0	0	0	9	74	216	268	177	49	6	0
	CDD18.3	44	0	0	0	0	0	0	11	26	7	0	0	0
	CDH23.3	51	0	0	0	0	0	1	18	23	8	0	0	0
	CDH26.7	3	0	0	0	0	0	0	1	1	1	0	0	0
Wind	WSAvg	6.8	8.1	7.8	7.4	6.7	6.0	5.7	5.6	5.5	6.3	6.9	7.4	8.1
Precipitation	PrecAvg	1500	148	131	117	112	101	104	88	100	119	153	153	172
	PrecMax	1851	227	182	221	187	177	173	170	228	308	267	241	355
	PrecMin	1151	70	48	47	59	49	51	37	44	43	61	96	97
	PrecStd	152	38	38	42	33	36	37	30	34	60	56	40	53
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	7.7	8.6	11.9	18.9	22.2	25.5	25.6	24.3	19.4	15.4	11.6
		MCWB	8.9	7.3	6.1	8.2	12.1	15.9	20.1	19.6	19.8	17.2	13.9	11.0
	2%	DB	7.1	5.6	6.4	9.7	15.3	19.7	23.2	23.7	22.0	17.7	13.6	9.7
		MCWB	6.7	5.0	5.2	7.0	10.9	14.6	18.6	19.2	19.2	16.2	12.6	8.9
	5%	DB	5.2	3.7	4.7	8.0	13.0	18.0	21.7	22.4	20.5	16.4	12.3	8.1
		MCWB	4.5	3.1	3.6	6.2	9.4	14.0	17.9	18.8	18.4	14.8	11.4	7.4
	10%	DB	3.4	2.2	3.2	6.6	11.2	16.4	20.5	21.4	19.5	15.1	11.0	6.4
		MCWB	2.6	1.5	2.0	5.0	8.6	13.5	17.5	18.5	17.5	13.6	10.0	5.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	7.5	7.2	10.1	13.7	18.0	21.1	21.8	21.5	18.2	14.4	11.2
		MCDB	9.3	7.8	7.8	10.9	17.2	20.0	23.1	23.2	22.9	18.6	14.9	11.4
	2%	WB	6.7	5.4	5.6	7.9	11.6	16.4	20.0	20.8	20.2	16.7	12.8	9.1
		MCDB	7.0	5.7	6.1	8.9	14.1	17.8	21.6	22.1	21.4	17.5	13.3	9.5
	5%	WB	4.7	3.3	3.9	6.6	10.2	15.2	19.2	20.1	19.0	15.1	11.5	7.5
		MCDB	5.1	3.7	4.5	7.6	12.1	16.7	20.6	21.4	20.2	16.1	12.0	8.0
	10%	WB	2.7	1.7	2.4	5.3	9.1	14.0	18.3	19.4	18.1	13.9	10.1	5.6
		MCDB	3.3	2.1	3.1	6.4	10.7	15.8	19.6	20.7	19.1	14.9	10.8	6.2
Mean Daily Temperature Range	5% DB	MDBR	6.5	5.9	5.2	5.2	6.2	6.3	6.0	5.2	5.2	4.9	5.0	5.4
		MCDBR	8.6	8.0	7.2	7.5	9.9	9.2	8.0	6.8	6.2	6.1	6.6	7.5
		MCWBR	9.1	8.1	6.5	5.7	5.9	5.4	4.5	3.9	4.3	5.6	6.6	7.9
	5% WB	MCDBR	8.6	7.8	7.1	6.9	8.6	7.7	6.7	5.4	5.6	5.9	6.3	7.5
		MCWBR	9.2	8.2	7.0	6.1	5.9	5.6	4.5	3.9	4.6	6.0	6.9	8.0
Clear-Sky Solar Irradiance	taub	0.303	0.297	0.323	0.357	0.374	0.393	0.422	0.401	0.361	0.339	0.322	0.301	
	taud	2.459	2.498	2.462	2.425	2.429	2.415	2.371	2.421	2.508	2.545	2.524	2.494	
	Ebn at Noon	811	895	912	904	895	875	844	850	860	834	783	775	
	Edh at Noon	70	80	96	108	112	114	118	108	91	77	66	61	
All-Sky Solar Radiation	RadAvg	1.23	2.12	3.40	4.22	5.00	5.34	5.29	4.84	3.88	2.47	1.38	0.94	
	RadStd	0.09	0.19	0.21	0.32	0.51	0.48	0.37	0.33	0.27	0.22	0.16	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+0.78	N/A	N/A	N/A	N/A	N/A	+25
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.43	N/A	N/A	N/A	N/A

Nomenclature: See separate page